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Design and Technical Specifications of the Integrated Conditioning and Acquisition Modules for MITICA Electrostatic Sensors

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ABSTRACT

This document presents the design and technical specification of the **Electrostatic Probe Integrated Conditioning and data Acquisition module (EPICA)**, a component of the MITICA Langmuir probe diagnostic. This new design takes advantage of the experience gained during the operation of electrostatic probes during SPIDER experimental campaigns [1], where the previous analog conditioning module [2] is still in use. These modules are designed to provide the specific analog signal conditioning, galvanic isolation and data acquisition interface between the probes in direct contact with the plasma and the MITICA CODAC system.

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1. Introduction

1.1. General introduction

This document describes in a self-consistent way the **Langmuir Probe Front End / DAQ**, part of the Electrostatic probes diagnostic system of the MITICA negative ion beam, under construction at the ITER NBTF facility located in the premises of Consorzio RFX in Padua, Italy.

This design structure system takes into account the experience gained with the system used in SPIDER [1][2] and the capabilities developed at Consorzio RFX at on the use of System on Chip (SOC, CPU+FPGA), System on Modules cards (SOM: SOC, RAM and I/O interfaces) boards [3].

It is worth recalling that this apparatus will operate in a hostile EMI environment, where continuous disturbance and random high-voltage spikes, due to the proximity of arc-discharges phenomena are constantly present during the measurements. This causes random electric potential differences between the reference ground of the probes and the ground of the conditioning/acquisition system, that if not properly handled, can not only hamper the measurements, but also damage the connected electronic systems, and in the worst cases even the probes and the vacuum feedthroughs.

The more traditional approach to solve the problem by using differential high impedance input presently does not appear a good choice. The main technical drawback of this method is its poor capability of rejection of high frequency disturbances (including those coming from the ubiquitous switched power supplies) and the need for additional protection components at the input, which degrade the measurements. Despite this and other limitations, this method was largely used in the past because of a combination of high cost and cumbersome complexity in realizing the galvanic isolation with conventional electronic components. In contrast, nowadays, dedicated components for high voltage isolation handling of signal and power control (ADCs, DC/DC converters, controllers, optical interfaces) combining competitive price and ease of use, are widely available on the market, driven by the volume production of photovoltaic and electrical automotive industry.

Thus at present, the most suitable method to keep the overall system safe and provide good quality measurements, consists in providing ***a galvanic isolation with a low stray capacitance toward the ground for each sensor***, either with respect to the conditioning/acquisition system local ground and among the sensor themselves.

1.2. Module definition for MITICA electrostatic probes electronics

Each module manages **16** separated **probes**, each with simultaneous voltage and current measurements for a total of 32 signals. Device configuration and data transfer are managed through Ethernet using TCP/IP and/or UDP protocol, in compliance with the general ITER CODAS specifications [4]. The timing can be managed either directly via Ethernet IEEE-1588 Precision Time Protocol following the indication of ref. [4] or alternatively by traditional external clock and trigger inputs, with an external timing decoder as needed.

The basic structure of the module is shown in Fig. 1.

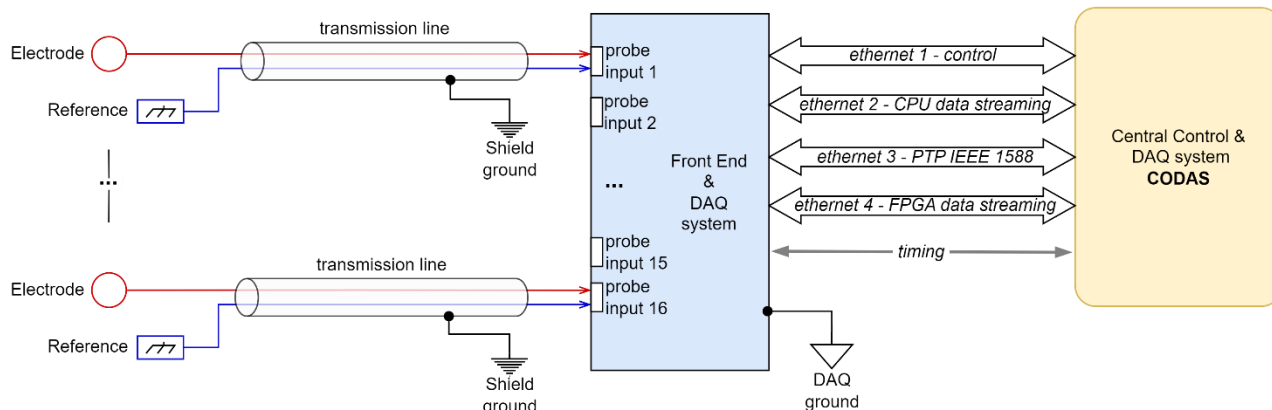


Fig. 1: Interface of the proposed module for electrostatic sensors with CODAS.

1.3. Specific requirements for MITICA

Three separate systems of electrostatic probes are present in MITICA [5][6], two integrated in the neutralizer and residual ion dump and a third one integrated in the calorimeter, for a total number of 212 sensors.

1.3.1. Neutralizer probes

In the Neutralizer, two Langmuir probe models are installed on the Front and the Rear panels, named respectively NEFES and NERES: NEFES consists of 6 electrodes and a ground reference signal for 4 panels and NERES of 16 electrodes and a ground reference signal for 2 panels. Collectively, their conditioning and acquisition can be managed using 4 modules for 62 probes.

1.3.2. Residual Ion Dump

In the Residual Ion Dump (RID), are present 6 electrodes on the top and 6 on the bottom bars for 4 panels: 48 probes, which can be processed by using 3 modules.

1.3.3. Calorimeter probes

In the Calorimeter, SED collects the secondary electrons emitted by the cooling tubes hit by the beam. They are organized in 3 horizontal arrays of 7 sensors for 4 panels and a vertical array of 18 sensors. The total number of 102 probes can be managed by 7 modules.

1.3.4. Number of conditioning/acquisition modules required

Taking into account the structure of the system and the distribution of probes in their respective sub-systems (see Tab. 1), the final requirement for the installation is for **14 modules**, plus a number of spare modules to be defined in the procurement contract.

MITICA component	Sensor type	Array structure	Num. of probes		Num modules (16 ch each)
Neutralizer	NEFES	(6 + 1) by 4	28	62	4
	NERES	(16 + 1) by 2	34		
RID	RES	6 by 2 by 4	48	48	3
Calorimeter	SED	18 + (7 by 3 by 4)	102	102	7
totals			212	212	14

Tab. 1: Number of signals and modules required.

2. Measurement types and configurations

The system implements the following type of measurements:

- Langmuir probe characteristic
- Single probe floating potential
- Single probe saturation current
- Double probe characteristic

2.1. The Langmuir probe characteristics

Two relevant parameters of the plasma (density and electron temperature) can be measured by means of Langmuir probes. The probes consist of sturdy electrodes immersed in the plasma and the relevant quantities are obtained by direct electrical measurement on the probe. For the purpose of the present specifications it suffices to recall the probes electrical I-V characteristic, somehow similar to that of a diode. Two example cases are reported in Fig. 2, in which the current is measured against a swept control voltage applied to the electrode. The measured currents can span from currents lower than the mA to several 100 mA range, depending on the plasma properties and the electrode geometry.

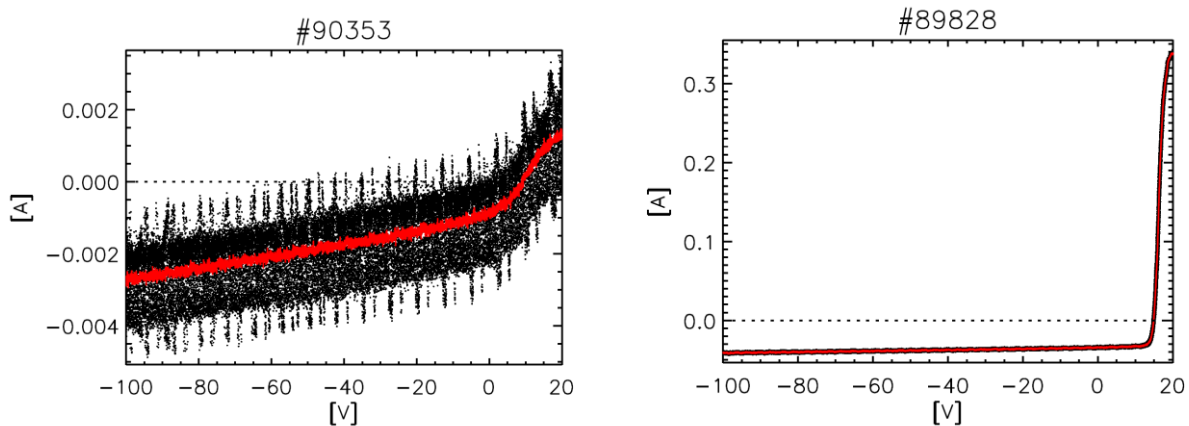


Fig. 2: experimental examples of Langmuir probe characteristics (from ref.[2]).

The basic setup circuit used to obtain the Langmuir probe characteristic, which is being implemented in the present system is shown in Fig. 3.

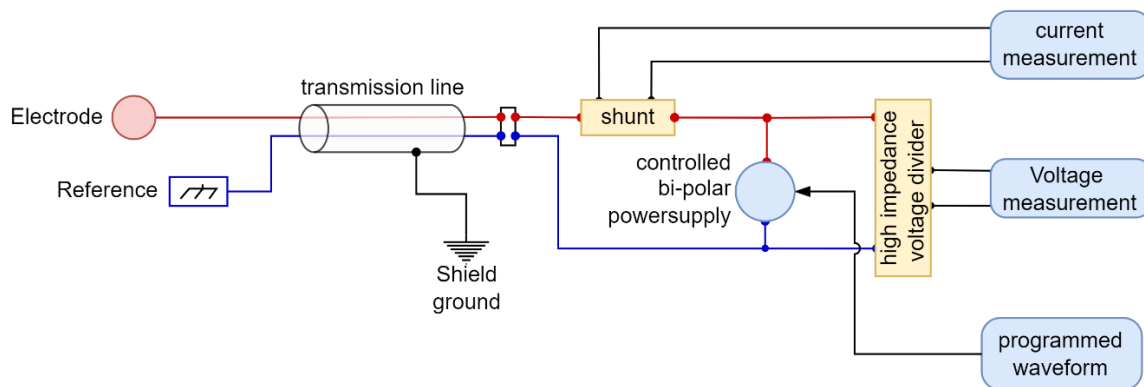


Fig. 3: schematic of a measurement setup for Langmuir probe characteristic.

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Fig. 4 shows other relevant quantities used with probe measurements:

- the *ion saturation current* (I_{sat}), which is the current measured at the maximum negative voltage allowed by the system (usually in the order of a negative -100 V);
- the *floating potential* (V_f), which is the potential of the probe at zero current, though in practical implementation it is connected to an high impedance input, accepting a small current leakage.

The electron saturation current is the saturation value of the electron branch of the I-V characteristic and depends on different factors. For the present purpose the current at the maximum meaningful voltage applied to the electrode is considered.

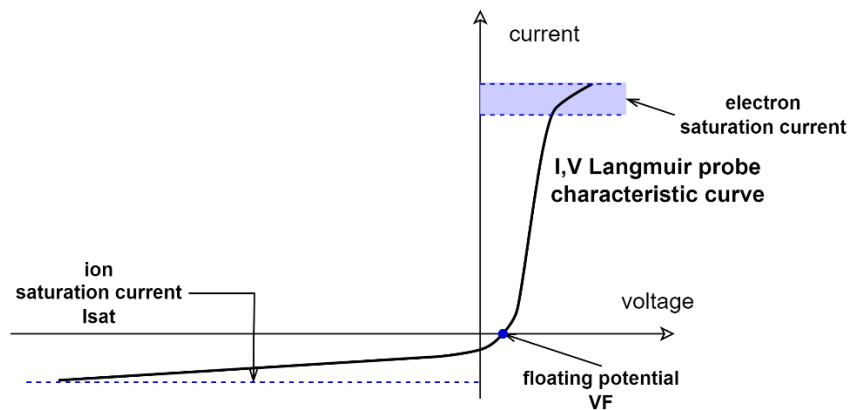


Fig. 4: relevant features of Langmuir probe characteristic: ion saturation current (I_{sat}), floating potential (V_f) and electron saturation current.

In the case of cold plasmas, the time required to sweep the voltage forth and back to obtain the characteristic can span from 10-20 ms when the plasma has high density, to some seconds required for low density.

However, many situations require also the measurements of plasma fluctuations, with higher time resolution, normally at least at some tens of kHz. In this case the characteristic is not feasible, so instead a continuous sampling of the I_{sat} or of the V_f is performed.

2.1.1. Floating Potential measurement - V_f .

The typical setup for the V_f measurement is the simplest and it is show in Fig. 5. While the absolute value of V_f depends on a variety of phenomena, its fluctuations are roughly proportional to those of plasma density and temperature. Even if the measure itself is not precise, it is often used given its simplicity and reliability.

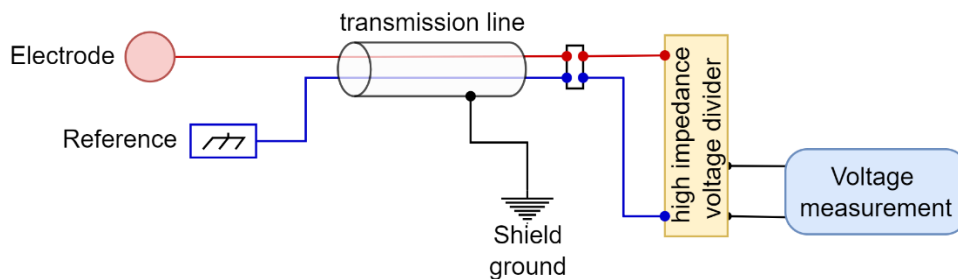


Fig. 5: schematic setup for practical floating potential (V_f) measurement.

Given that an excessive electron saturation current can damage the probe electrode and V_f can be either positive or negative, its value becomes an important reference to determine a safe voltage range for the characteristic.

2.1.2. Ion saturation current measurement - I_{sat}

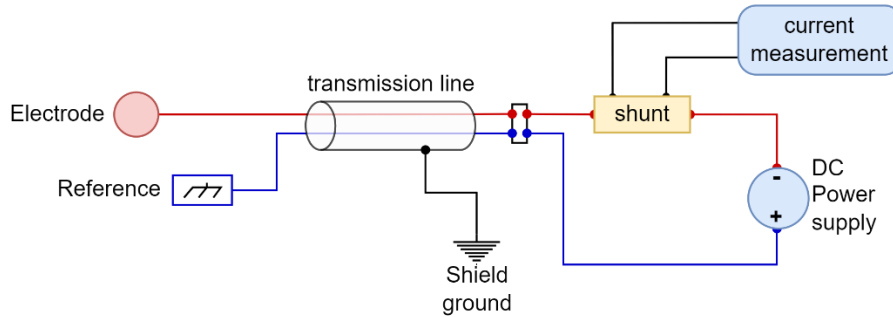


Fig. 6: schematic setup for ion saturation current (I_{sat}) measurement.

The ion saturation current measurement provides a good estimate of the plasma density. It requires only a constant voltage DC power supply (note the probe has a negative polarization) and a current measurement (Fig. 6). Of course, the single probe configuration can be used as a current saturation measurement by simply forcing the variable power supply at a full negative constant voltage and discarding the voltage measurement. The advantage of this configuration is its ability to give a measurement of plasma density fluctuations with high time resolution.

2.1.3. Double probe measurement

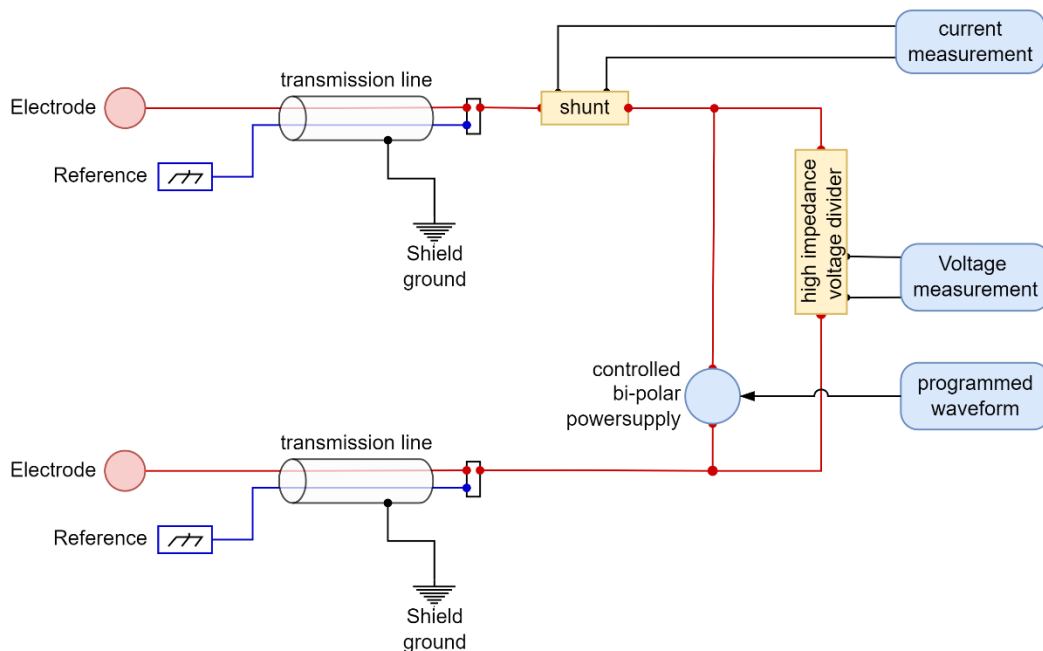


Fig. 7: schematic setup for double probe measurement.

The double probe setup is shown in Fig. 7 and its resulting characteristic curve in Fig. 8. Due to the arrangement of the probes, it provides a reliable setup because of the intrinsic self-limitation of the current. Both plasma density and temperature can be measured without incurring in the problems linked to electron saturation current.

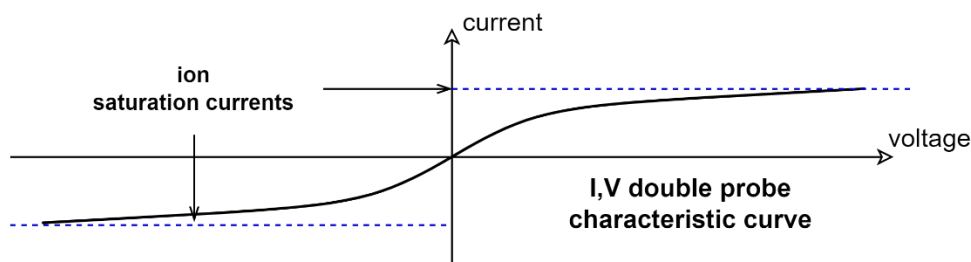


Fig. 8: double probe current-voltage characteristic.

However, this setup is not easy to realize, mainly because of its sensitivity to current leakages, especially in poorly arranged setups. So when using this setup, floating power supply and I-V measurements with good galvanic isolation, become essential features to minimize the current leakages.

3. General System description

This chapter gives a general specification for the appliance. The detailed specification for the measurements and the power supply for probe polarization are discussed in the subsequent chapters.

The proposed system is designed as a self-contained module composed by 8 units, each unit managing **one power supply shared between two separated probes**. Practically the system is capable to manage up to **16 probes with 8 power supplies**. These numbers allow sufficient flexibility for the system, while limiting cost and power requirements. By using 8 separated power supplies instead of 16, keeps the system compact and less expensive, because in most cases pairs of adjacent probes can share the same ground reference.

Yet the few special cases where a single probe must be individually handled, it can be managed by disabling one of the two channels.

The system shall be managed by a commercial SOC system. A reasonable choice for this application could be the [Kria SOM](#), recently introduced by Xilinx.

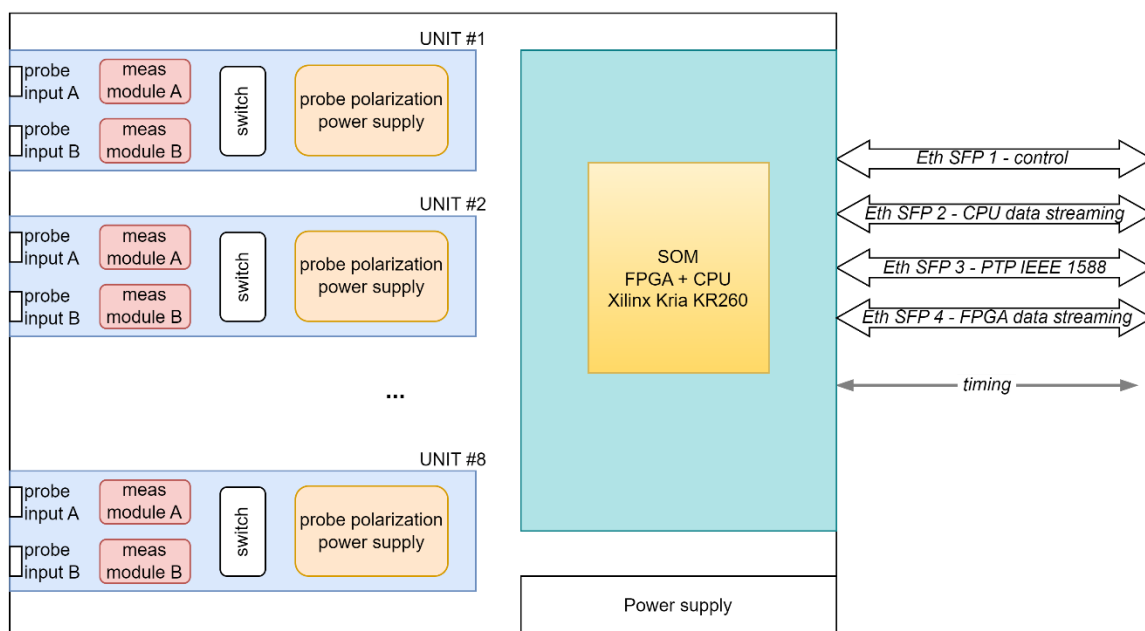


Fig. 9: general layout for an 8 unit / 16 probes module.

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3.1. Data Interface

The system shall provide four separated ethernet interfaces, made available through SFP+ connectors.

The purpose of the interfaces follows the recommendation for the ITER network [4]:

- a) Device control and management
- b) Bulk diagnostic data transmission
- c) Timing, through PTP/IEEE 1588
- d) Real time data transmission

The interface d) shall be directly connected to one of the GTH transceiver ports of the FPGA, so it can be configured for transmitting data via different standards such as ethernet UDP datagrams, PCIe or custom serial high-speed link.

Additionally, a micro-SD holder must be accessible, along with a USB port configured as TTY console access on the control unit for software updates and debugging purposes.

3.2. Trigger and timing interface

The timing interface will provide two optical SC inputs and two Lemo 00/Fischer compatible connectors.

The requirement for a basic interface is to use the two inputs in the conventional way: one for clock input/synchronization and one as trigger input.

These inputs are to be routed into the FPGA, leaving the possibility to redefine their capability for handling different synchronization standards like MPB [7] or ASDEX/W7-X [8].

In addition, one of the SFP+ Ethernet ports has to be assigned to allow the system to properly manage data time stamping using the IEEE-1588 Precision Time Protocol (PTP) as an absolute timing reference.

3.3. Power supply

The basic specification for the power supply is a universal AC (110-240V, 50 or 60 Hz). An additional separate screw connector for the shielding ground shall be made available on the rear panel.

Possible optional additional features could include:

- implementation of a "Wake-on-LAN" mechanism [9].
- power supply from a 24 VDC source, for example to be used with battery.

3.4. Dimensions

The module dimensions shall be contained within a 3.5 U or less standard rack module height. The maximum allowed depth is 500 mm.

3.5. Front and rear panels

Except for the power supply on the rear, all connectors and interfaces shall be placed on the front panel.

3.6. Environment

The module is meant to be operated in dedicated cubicles. Typical operation temperature between 20 and 40 °C, nevertheless the device shall be capable to regularly operate between 0 and 50 °C, with non-condensing humidity.

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4. Two channels unit description

As described above, each module is composed of eight fully independent units. Each unit contains two measuring modules, one switch module for the configuration of the measurements and a power supply (Fig. 10).

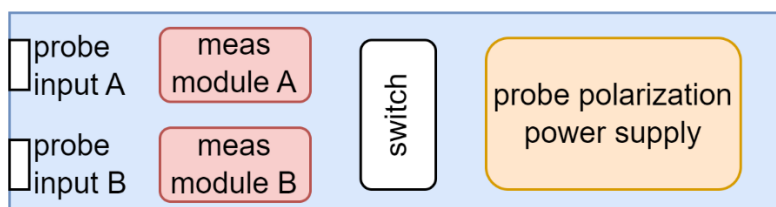


Fig. 10: two channel unit structure.

4.1. Measurement modules

Each probe input measurement module (Fig. 11) consists of two isolated ADCs, for the simultaneous sampling of the applied voltage and of the current flowing into each probe (Fig. 11). In order to simplify the design, while maintaining adequate performance, a possible good solution can be the use of isolated Σ - Δ ADCs for automotive application (e.g. [ADUM77xx from Analog Devices](#), [AMCxxxx from Texas Instruments](#), [ISOSD61 from STMicroelectronics](#)), which provides an equivalent 16 bit 50 kSamples/sec ADC, with a single bit data interface. The sampling frequency can be increased up to 100 kS/sec, reducing the precision to 14 bit.

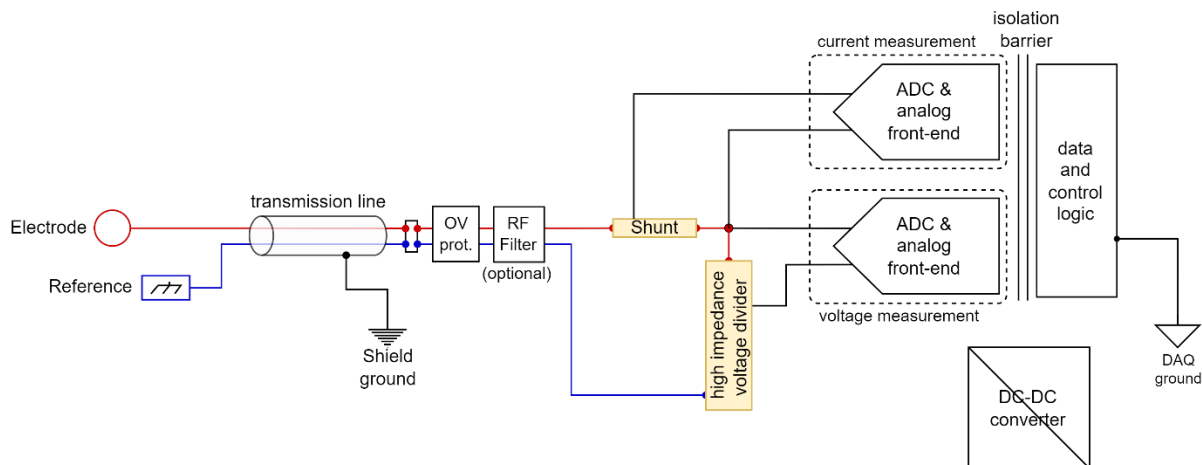


Fig. 11: schematic layout of an input probe module

The range and nominal resolution requirements for the measurement are reported in Tab. 2.

At the input a receptacle for hosting an optional customizable RF filter module mounted on a separated board, shall be provided (Fig. 12). In case the filter would not be necessary, it shall be replaced by a dummy board providing a straight connection.

Moreover, a suitable overvoltage protection shall be provided for each input.

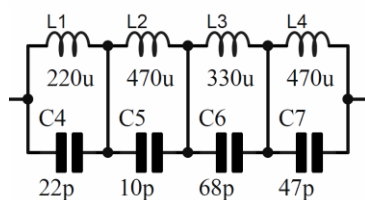


Fig. 12: schematic of a possible 4 tap 1 MHz RF notch filter.

Measurement Range	Minimum nominal resolution	Maximum residual r.m.s. noise
Voltage measurements		
± 120 V	50 mV	15 mV
Current measurements		
± 1 mA	0.5 μ A	0.15 μ A
± 10 mA	5 μ A	1.5 μ A
± 100 mA	50 μ A	1.5 μ A
± 1 A	0.5 mA	0.15 mA

Tab. 2: measurement ranges and relative resolution requirements.

The instrument shall be capable of self-calibration so that the overall precision of the measurements are maintained within 1% for the nominal values at least 5 years.

4.2. Power supply module for probe polarization

The usual three coarse requirements for the choice of a suitable power supply for Langmuir probe measurements, are:

- Maximum voltage
- Maximum current
- Presence of an active power component - the probe behaves as a voltage generator.

When straightforwardly applied, looking for a commercial off the shelf solution, these requirements quickly lead to over specification (Fig. 13).

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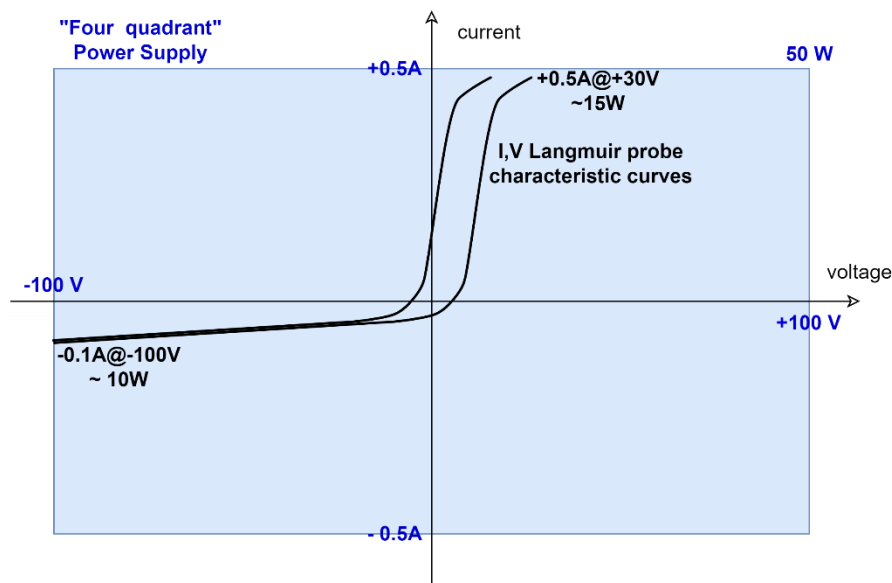


Fig. 13: power level needed when using commercial “4 quadrant” power supplies for I-V Langmuir characteristic.

For example standard industrial power supplies used for testing/driving reactive systems with inductive and capacitive loads, fully satisfy the three requirements above. These units are usually based on traditional class AB or class G bipolar linear amplifiers. Even if rather complex, bulk and consequently relatively expensive, they nevertheless provide a reliable and straightforward solution to the requirements. While this choice is the most rational one for practical purpose when dealing with a limited number of probes, it becomes hardly viable when applied on a large number of probes (say more than 50). On the other hand, once the number is sufficiently large as in this case, the development cost of a bespoke solution becomes viable, technically more efficient and cheaper.

By carefully analyzing the requirements, it turns out that some relevant features of test amplifiers are mostly useless in this application. Looking at the Langmuir characteristic:

- The maximum current is needed at relatively low voltage and the maximum voltage at comparable low current: in principle the effective required power is less than one third of that nominally required.
- The “active” component of the probe occupies a small fraction of I-V diagram around the zero, with low voltage *and* current requirements, a situation that can be simply managed with a ballast resistor (details are discussed later on).
- Since the characteristic is not strictly time dependent, the harmonic distortion of a switching glitch at the zero crossing is not an issue in this context.

“Net” diagrams need for efficient operation are shown in Fig. 14 and Fig. 15.

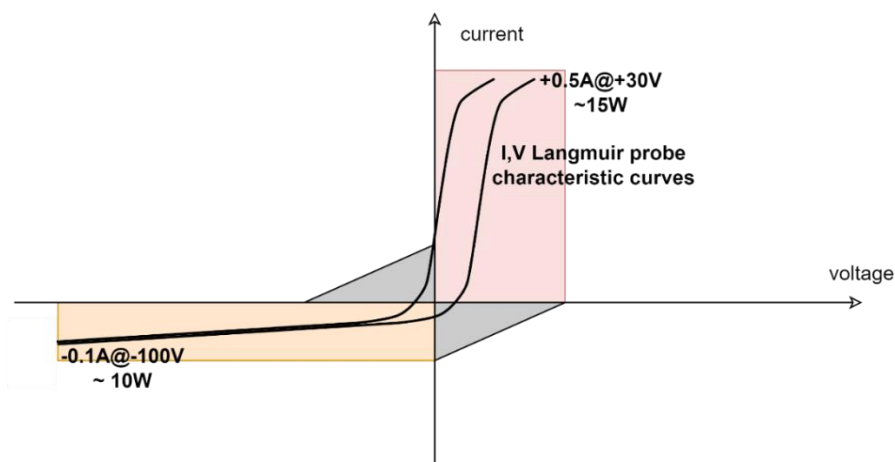


Fig. 14: “net” operating space I-V diagram and power requirements for Langmuir probe characteristics.

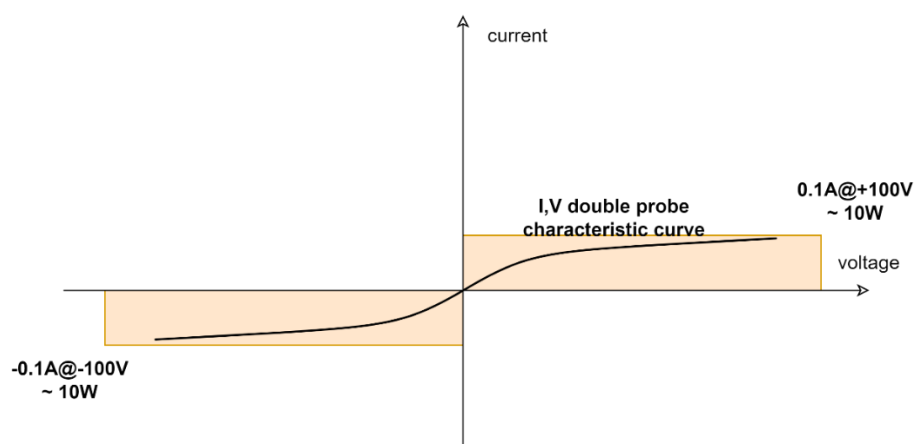


Fig. 15: “net” operating space I-V diagram and power requirements for double probe measurement.

From the dynamical point of view, in this type of application the maximum expected repetition rate can be up to 100 sweeps per second. This roughly translates in the need of a triangular wave with a frequency of up to 50 Hz, which gives two sweeps for each up-down cycle. Because of the harmonic content of a triangular wave at 50 Hz, an acceptable one requires an equivalent analog bandwidth of at least 1 kHz, even if likely the system will be mostly operated at lower frequencies. Nevertheless, since the final voltage waveform could have different shapes, depending on the operational requirements and should be freely programmable with any arbitrary shape, the 1 kHz bandwidth adequately fulfills these further requirements.

A possible implementation can be realized with the scheme shown on Fig. 16. This topology is a combination of a switching layout borrowed by H-bridges used by power inverters, with two linearly controlled single polarity power supplies. One power supply for low voltage high current (**0-30V, 0.3A**) and one for higher voltage with lower current (**0-100V, 0.1A**). The voltage need for the I-V characteristic can be configured by proper control of the various switches, practically implemented by MOSFETs or IGBTs. Nevertheless, other implementation schemes are also possible.

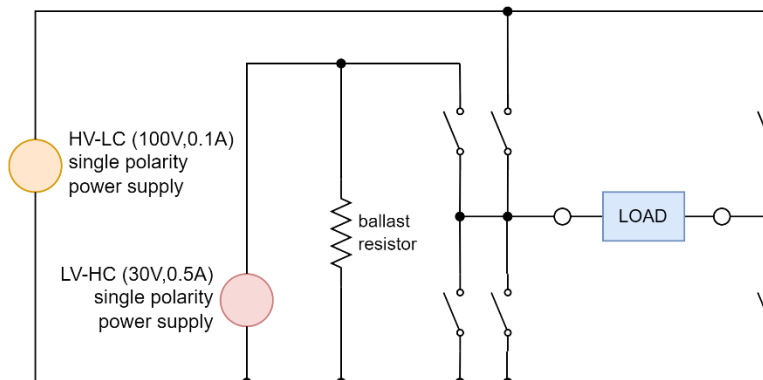


Fig. 16: a possible efficient power supply layout scheme.

4.3. Configuration switching modules

The configuration switch manages the connection between the inputs, the measurement modules and the power supply. Each probe can be configured for the different types of measurement:

1. Floating Potential - VF, i.e. without biasing.
2. with probe biasing - IS/SLP, i.e. in Ion Saturation current or Single Langmuir Probe.
3. in Double Probe.

This gives five different types of measurement configurations requested for the unit, as reported on Tab. 3.

	Probe A	Probe B	Power supply
1	VF	VF	disconnected
2	SLP / IS	VF	Probe A
3	VF	SLP / IS	Probe B
4	SLP / IS	SLP / IS	A parallel B
5	Double probe		A series B

Tab. 3: measurement configurations for each unit.

Given the fact that a single biasing power supply is present, there would be some limitations on using two probes in parallel:

- The two probes must physically share the same ground reference, or at least two very close each other.
- The maximum available current from the biasing must be shared between the two probes, so the maximum measurable density is roughly halved.

Nevertheless, in most of the situations a proper installation can avoid the first limitation, while the second case is quite unlikely for the typical expected plasmas. In any case they can be circumvented by performing alternate measurements by switching between configuration 2 and 3.

A possible implementation scheme satisfying all the required configuration is reported in Fig. 17. Note the presence of two resettable fuses, which are intended for protection of configuration 4, avoiding excess current in case of accidental connections creating a voltage difference between the references of probe A and B.

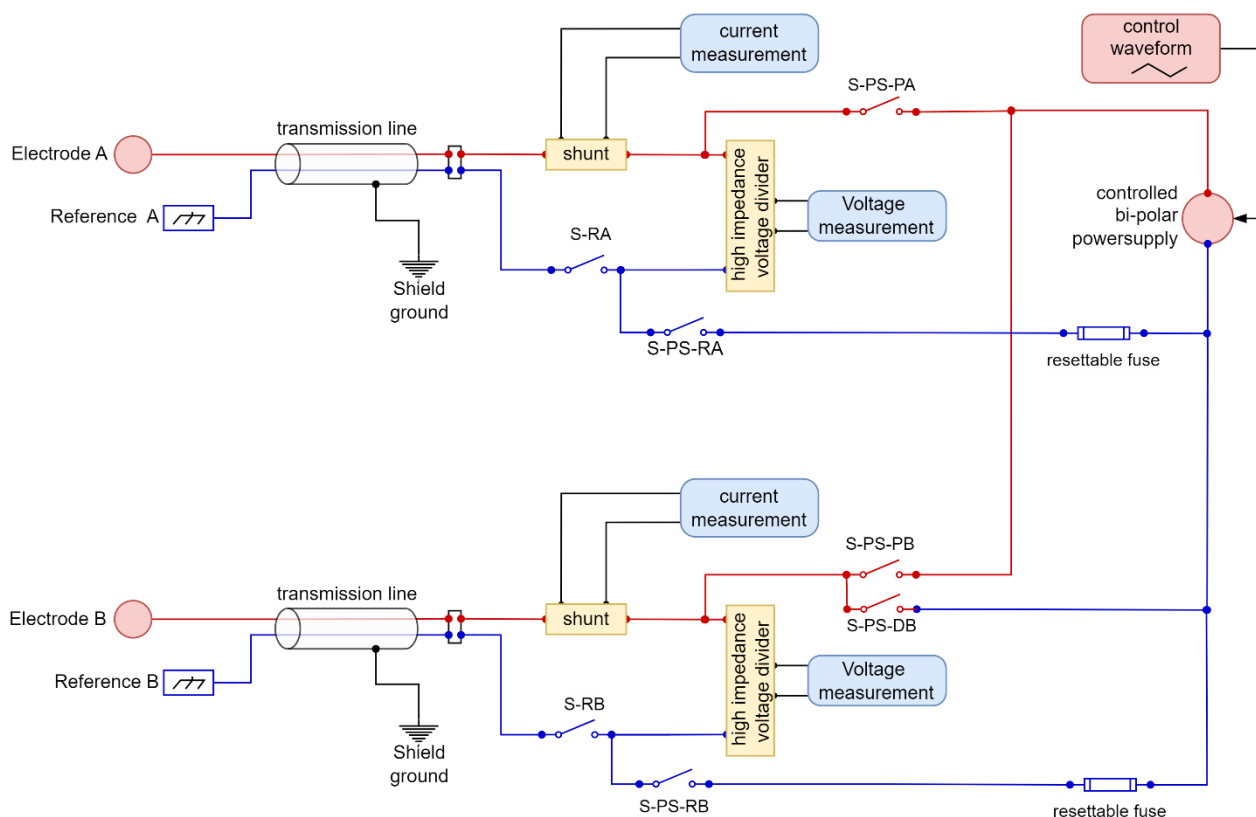


Fig. 17: schematic of the switches connecting the probe A and B, with the biasing power supply.

The coding for the switches can be intended as:

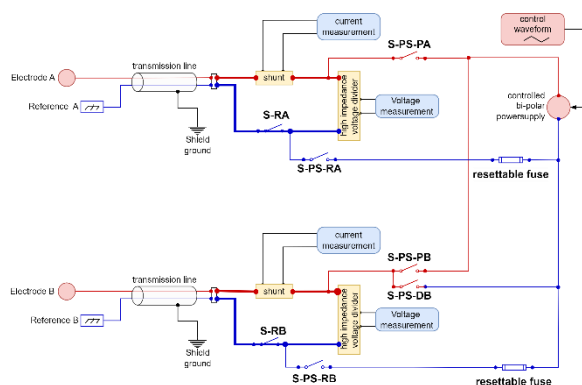
- S: switch
- PS: connection for Power Supply
- PA, PB: connection for Probe A and B
- RA, RB: connection for Reference of A and B
- DB: connection for Double probe operation, with “cold” connection on electrode B

By using this scheme, the different measurement configurations can be obtained by means of the connections reported in Tab. 4.

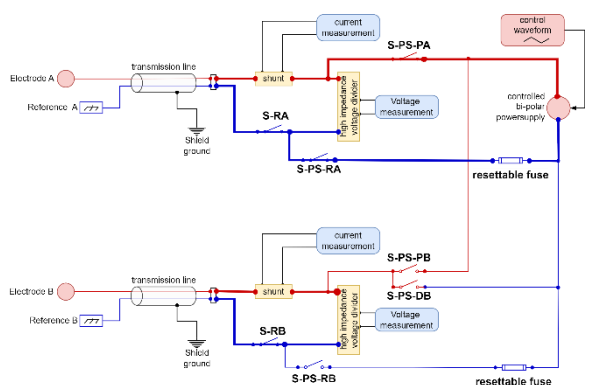
Cfg	Probe A	Probe B	S- PS-PA	S- PS-RA	S- RA	S- PS-PB	S- PS-DB	S- PS-RB	S- RB
1	VF	VF	open	open	close	open	open	open	close
2	SLP / IS	VF	close	close	close	open	open	open	close
3	VF	SLP / IS	open	open	close	close	open	close	close
4	SLP / IS	SLP / IS	close	close	close	close	open	close	close
5	Double probe		close	close	open	open	close	open	open

Tab. 4: switch configuration table

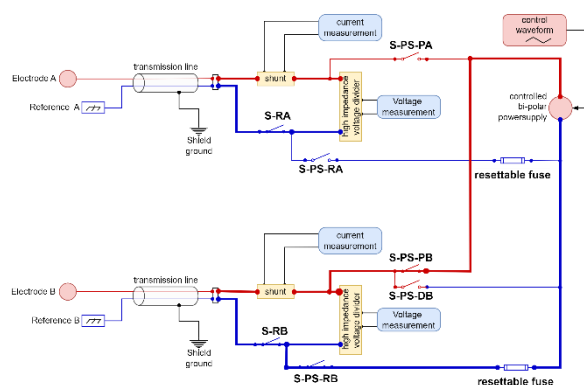
For the sake of clarity, the circuit paths used in each configuration are highlighted on Tab. 5.



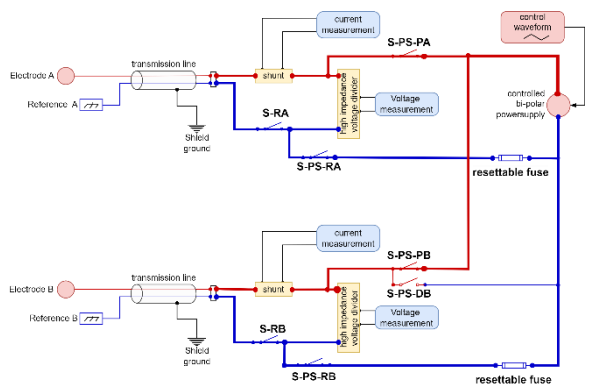
Configuration 1: Probe A: VF – Probe B: VF



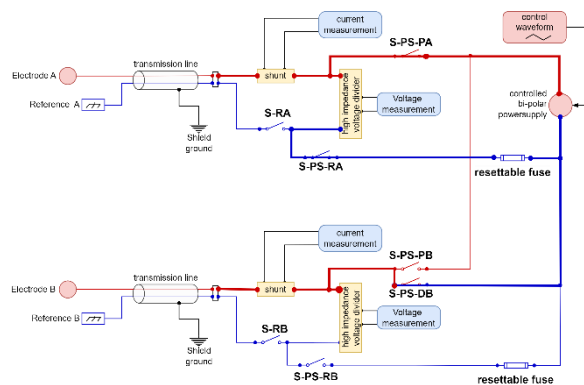
Configuration 2: Probe A: IS/LPC – Probe B: VF



Configuration 3: Probe A: VF– Probe B: IS/LPC



Configuration 4: Probe A: IS/LPC– Probe B: IS/LPC



Configuration 5: Double Probe

Tab. 5: connection paths of each configuration (refer to Fig. 17 and Tab. 4)

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5. Software requirements for module management

The system is intended to be operating using a Linux operating system recommended by the SOM manufacturer, usually Ubuntu or PetaLinux.

The system must be well documented by the supplier, with all the related hardware schematics, VIVADO files and operating system images.

The requested software from the manufacturer are the firmware development for the FPGA, the device drivers and documented APIs (application Program Interfaces).

For the supply, only a basic implementation is required, which provides a set of suitable simple textual SCPI commands, for testing requirements.

Advanced management and real-time control will be later implemented outside this supply, using a MARTE server [10][11], deployed by Consorzio RFX, directly running on the SOM module.

The MARTE application will supervise the device and will present itself to the external world via the network (TCP/IP for commands, UDP for data streaming). No other ways for communication between the central Control and Data Acquisition system are foreseen except the network connection. The MARTE application shall also be responsible for the real-time generation of the reference signals to the power supplies and of the data readout, receiving commands and sending data from/to the network and interacting with the Device driver interfacing the FPGA. A possible option for faster data throughput is a UDP communication link directly managed by the FPGA, which could be developed in the future if it is deemed necessary.